Solvency-Based Pricing & Stress Testing under EV Transition Scenarios

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Executive Summary

This project develops a **solvency-based pricing framework** for motor third-party liability insurance that explicitly incorporates **tail risk and capital costs** into premium determination. Rather than relying solely on expected loss, the analysis models the **full distribution of aggregate portfolio losses** using a frequency–severity structure combined with Monte Carlo simulation, allowing capital requirements and risk load to be quantified in a coherent manner.

Using the French Motor Third-Party Liability (MTPL) dataset covering **413,169 policies**, claim frequency is modeled at the policy level using a **Negative Binomial generalized linear model (GLM)** to account for observed over-dispersion and unobserved heterogeneity. Claim severity, conditional on occurrence, is modeled using a **Lognormal generalized linear model**, reflecting t the heavy-tailed behavior observed in the data.

Aggregate annual losses are simulated through a compound frequency–severity framework. From the simulated loss distribution, we mainly use Expected Loss (EL)**,** Value-at-Risk (VaR)**,** and Expected Shortfall (ES) to **estimate** at the portfolio level. A **capital-based pricing rule** is then applied, where the premium consists of expected loss plus a risk load proportional to the cost of holding capital against extreme loss scenarios. The cost of capital is calibrated using an external industry benchmark, resulting in an illustrative premium that reflects both average risk and tail uncertainty.

To assess pricing robustness under structural risk changes, the framework is extended to an **electric vehicle (EV) transition stress test**. Rather than attempting to predict future EV adoption, the analysis adopts a **stress-testing perspective**, applying plausible shocks to claim frequency and severity motivated by industry evidence. These scenarios are designed to examine how shifts in loss distribution shape affect capital requirements and risk-adjusted pricing.

The stress test results highlight that, under certain assumptions, reductions in claim frequency may have more influence on premium than increases in severity, leading to a lower point estimate of premium. However, this does not imply a reduction in long-term risk. Instead, the findings illustrate how **capital-based pricing frameworks calibrated primarily on mean adjustments may fail to capture tail risk, parameter uncertainty, and model risk during periods of structural transition**.

Overall, the project demonstrates the value of **distribution-aware pricing and stress testing** in insurance risk management. By linking pricing directly to capital requirements and tail behavior, the framework provides a transparent and extensible tool for evaluating solvency, robustness, and strategic risk exposure under emerging risks such as EV transition.

Introduction

Pricing under uncertainty requires more than estimating expected outcomes. In insurance and other risk-transfer mechanisms, losses are inherently stochastic and often exhibit heavy-tailed behavior, making extreme but infrequent events a key driver of financial risk. As a result, pricing based solely on expected loss fails to account for the capital required to absorb adverse scenarios.

This project develops a risk-aware pricing framework that explicitly links loss distribution characteristics to premium determination. Using a frequency–severity modeling approach combined with Monte Carlo simulation, the analysis constructs the annual aggregate loss distribution of a large portfolio and quantifies tail risk through measures such as Value-at-Risk and Expected Shortfall. The difference between these tail risk measures and expected loss is interpreted as a **risk load**, representing the price of uncertainty beyond the mean.

To assess the robustness of pricing under structural risk changes, the framework is further subjected to stress scenarios motivated by electric vehicle (EV) transition risks.
Rather than forecasting future outcomes, these scenarios perturb the underlying loss distribution to examine how sensitive risk loads and capital requirements are to shifts in frequency, severity, and dispersion under the revolution of car industry. The results illustrate how risk-adjusted pricing responds to changes in tail behavior and highlight the trade-off between competitiveness and solvency.

Data & Scope

Data Summary

The analysis is based on the *French Motor Third-Party Liability (MTPL)* insurance dataset, publicly available on Kaggle and originally compiled for actuarial modeling research. The dataset contains information on **413,169 motor insurance policies**, observed predominantly over a one-year period.

Two related datasets are used. The *frequency dataset* includes policy-level exposure information, claim counts, and structured risk characteristics such as driver age, vehicle age, geographic density, and power group. The *severity dataset* records individual claim amounts corresponding to reported claims. The two datasets are linked through a common policy identifier.

The data is largely pre-processed and does not contain missing values in the variables used for modeling. As a result, no additional data cleaning procedures were required beyond variable selection and basic consistency checks.

Scope of Analysis

The scope of this study is confined to **risk-based pricing and capital assessment at the portfolio level** over a one-year horizon.

Losses are modeled on an annual basis, consistent with standard insurance pricing and solvency frameworks.

The analysis focuses on structured rating variables commonly used in technical pricing models. Individual-level underwriting considerations, such as credit history or discretionary accept decline rules, are not included, as these are typically addressed outside actuarial pricing models.

The risk framework considers insurance risk arising from claim frequency and severity uncertainty. Other sources of risk, including market, credit, or operational risk, are not modeled.

Finally, EV transition scenarios are introduced as **stress tests** to examine the robustness of pricing under structural changes in loss behavior. These scenarios are not intended as forecasts of future adoption rates but as perturbations to the underlying loss distribution.

Methodology

1.Frequency Model

1.1Modelling Objective

The objective of the frequency model is to estimate the **policy-level expected number of claims** over a one-year horizon, conditional on observable risk characteristics.
Rather than predicting realized claim counts, the model focuses on estimating the conditional mean of claim frequency, which serves as a fundamental input to aggregate loss simulation and risk-based pricing.

1.2Data Description & Exploratory Analysis

This study models annual claim frequency using the *freMTPLfreq* portfolio, which contains **413,169** motor third-party liability policies observed primarily over a one-year period. The response variable is the **policy-level claim count** Ni (denoted ClaimNb), accompanied by exposure information (Exposure) and structured rating variables (e.g., driver age band, vehicle age, geographic density, vehicle power group, brand, fuel type, region).

It's shown below the summary of the estimated claim number for every policy:

Variable	**Mean**	**Variance**
ClaimNb	0.03916315115606447	0.04163754465805693

Table1. Mean & Variance of claim number

Figure 1. below shows the empirical distribution of ClaimNb. The distribution is highly **right-skewed** with a large mass at zero (most policies experience no claims over the observation window) and rapidly decreasing frequency for multiple-claim outcomes. Summary statistics for key variables are provided in **Appendix A**.

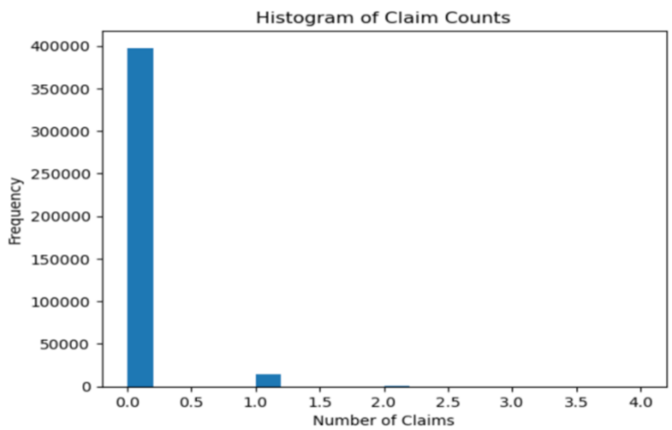

Figure1.Distribution of claim numbers

Exploratory analysis shows that diesel vehicles exhibit a slightly higher empirical claim frequency than regular gasoline vehicles, with a balanced exposure split across fuel types. However, gas type is excluded from the baseline frequency model to maintain a consistent and interpretable factor structure when extending the analysis to electric vehicle (EV) scenarios, for which gas type is not defined. As the focus of the study is on stress testing under EV transition scenarios, excluding gas type avoids introducing structural inconsistencies in subsequent model extensions.

1.3 Variable Grouping and Risk Segmentation

Prior to model estimation, key rating factors were grouped into actuarially meaningful bands to capture non-linear risk patterns and ensure statistical credibility. Several continuous variables, such as driver age, vehicle age, and population density, exhibit highly skewed distributions and non-linear relationships with claim frequency, making a linear specification inappropriate.

Grouping was therefore applied based on a combination of domain knowledge and exposure-weighted diagnostics, allowing the model to reflect structural risk differences while avoiding unstable estimates driven by sparsely populated segments. High-cardinality categorical variables with limited exposure were consolidated to improve parameter stability and interpretability. More detailed information about grouping is provided in Appendix A.

This approach aligns with standard actuarial pricing practice, where risk segmentation is performed prior to parametric modeling to balance model flexibility with robustness. We then attempt to find the general linear relationship between Claim number and DriverAge, CarAge, geographical density, car power type, brand, region.

1.4 Baseline Poisson Model and Distributional Assumptions

As a natural starting point for modeling claim frequency, a Poisson generalized linear model (GLM) is considered. Claim counts are non-negative integers observed over a fixed exposure period, making the Poisson distribution a standard baseline in frequency modeling.

Under a Poisson assumption, the conditional distribution of claim counts satisfies：

$$N_i|X_i \sim Poisson(\mu_i), E[N_i|X_i] = \mu_i, Var[N_i|X_i] = \mu_i$$

where μ_i denotes the expected number of claims for policy i and X_i implies the array of predictor factors. The model links the conditional mean to rating factors via a log link function,

$$\log(\mu_i) = \beta_0 + \beta^\tau X_i + \log(e_i)$$

with $\log(e_i)$ included as an offset to account for varying exposure across policies. This specification implies that rating factors act multiplicatively on expected claim frequency, which is consistent with actuarial interpretations of relative risk.

We then build a full model with all parameters: Driver Age, Car Age, Geographical Density, Car Power Type, Brand and Region. And we try to delete the comparable insignificant factor Region and find AIC get larger. Then we decide to reserve the factor for the better interpretation of the model.

Model	**AIC**
Full Model	135337.94287078088
Simplified Model without Region	135347.80471455926

Table2.AIC of different model

Also, we noticed that empirical diagnostics based on the observed claim counts indicate that the variance of claim frequency substantially exceeds the mean, providing clear evidence of **overdispersion** relative to a Poisson assumption. This suggests that the Poisson specification may underestimate variability in claim occurrence and, by extension, tail risk at the portfolio level.

1.5 Negative Binomial Specification

To address overdispersion, the frequency model is extended to a **Negative Binomial (NB) GLM**, which retains the same conditional mean structure as the Poisson model but allows for more flexible variance behavior. Specifically, the NB model assumes:

$$Var(N_i|X_i) = \mu_i + \alpha \cdot \mu_i^2, \quad \alpha = (Var(N_i) - Mean(N_i))/Mean(N_i)^2$$

where $\alpha > 0$ is a dispersion parameter capturing excess variability beyond the Poisson benchmark. When $\alpha \to 0$, the model collapses to the Poisson case.

When building a NB model, we find the aic of the modified model is smaller than Poisson model, indicating a better fit.

Model	**AIC**
Poisson GLM	135337.94287078088
Negative Binomial GLM	134976.77225403092

Table3.AIC of model using different distribution

From a risk modeling perspective, the dispersion parameter α can be interpreted as reflecting **unobserved heterogeneity** in underlying risk levels across policies. Even after conditioning on observed rating factors, individual policyholders may differ in latent risk characteristics, leading to greater variability in claim counts than implied by a homogeneous Poisson process.

The Negative Binomial specification therefore provides a more realistic representation of claim frequency in the presence of heterogeneous risks. Importantly, it preserves interpretability of coefficient estimates while improving robustness of downstream risk aggregation. Fitted values from the NB model yield policy-level expected claim frequencies $\widehat{\mu_i}$, which are subsequently used as inputs to Monte Carlo simulation of aggregate portfolio losses.

1.6 Summary

Estimated coefficients exhibit economically intuitive patterns across key rating factors such as driver age, vehicle characteristics, and geographic density. From the model, it showed that higher population density is associated with increased claim frequency, while middle-aged drivers exhibit lower expected claim counts relative to younger cohorts. For completeness, full model outputs are reported in Appendix B.

2.Severity Model

2.1 Modeling Objective and Data Structure

Claim severity is modeled conditional on claim occurrence and focuses on the distribution of individual claim amounts. Unlike frequency, which is defined at the policy level, severity is observed only for policies with positive claims and is therefore modeled at the claim level.

$$X = ClaimAmount \mid ClaimAmount > 0$$

The empirical distribution of claim amounts exhibits strong right skewness, with a small number of large losses contributing disproportionately to total loss. All zero-loss records are excluded from severity modeling to avoid mixing frequency and severity effects.

2.2 Distributional Choice for Severity

Figure 2 shows the histogram of Claim Amount on the original scale and on the log scale. The raw-scale distribution is strongly right-skewed with a long right tail, while the log-scale histogram is substantially more symmetric. In addition, a small number of large claims dominate tail risk, which is quite important when we dive into the tail behaviours.

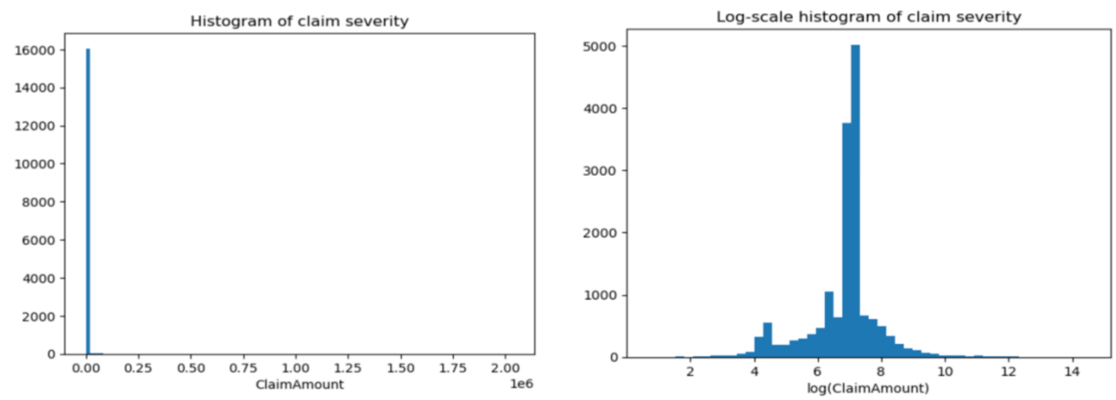

Figure2.Distribution of claim amount and log (claim amount)

Exploratory analysis indicates that, compared to claim frequency, severity shows weaker and less stable dependence on available policy-level rating factors such as driver age, vehicle age, and density bands. For example, figure3 illustrates claim severity by driver age band on a logarithmic scale. While the distribution exhibits substantial right-skewness and heavy tails across all age groups, differences in central tendency between age bands are relatively small compared to the overall within-band dispersion.

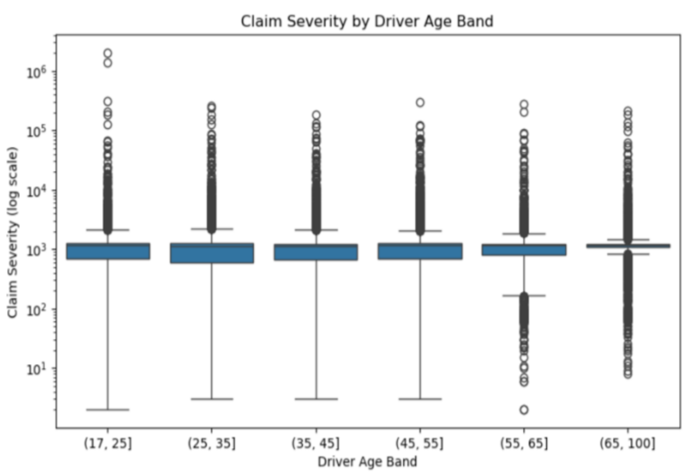

Figure3.Claim Amount over different Driver age

2. Only Intercept Lognormal Severity Model

Given the limited explanatory power of policy-level characteristics for claim severity and the heavy right-skewed nature of claim amounts, a **parametric lognormal specification** is adopted as a parsimonious baseline severity model. Specifically, for positive claim amounts X > 0, the following assumption is made:

$$\log(X) \sim \mathrm{N}(\mu, \sigma^2)$$

where μ and σ^2 are estimated using ordinary least squares applied to log-transformed claim amounts. This approach captures the strong skewness and heavy-tailed behavior observed in the empirical severity distribution while avoiding over-parameterization in the absence of robust severity predictors.

Diagnostics (figure 4) reveal that the claim severity is heavy-tailed. The upward deviation from the red line suggests that extreme losses occur more frequently or with higher severity than a standard Lognormal distribution predicts.

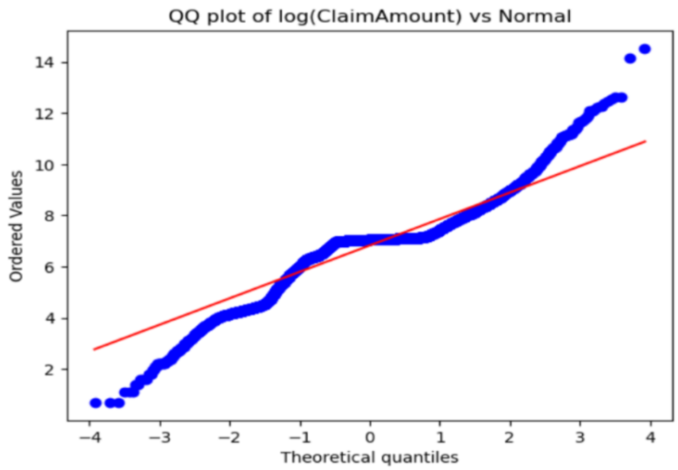

Figure 4.Q-Q Plot of log(amount) vs. Normal Distribution

3.Monte Carlo Aggregation

3.1 Compound Loss Framework

Portfolio loss is modeled using a **compound frequency–severity structure**. For a portfolio of policies i = 1,2,3… the annual total loss is:

$$L = \sum_{i=1}^{m} \sum_{k=1}^{Ni} X_{ik}$$

where:

- N_i is the claim count for policy i(frequency),
- X_{ik} are i.i.d. severities drawn from the severity distribution (conditional on a claim occurring).

3.2 Model Used in Simulation

For frequency, since claim counts exhibit over-dispersion relative to Poisson (variance exceeds mean). Therefore, frequency is simulated using a Negative Binomial (NB) model where:

$$N_i \sim NB(\mu_i, \alpha), E[N_i] = \mu_i, Var(N_i) = \mu_i + \alpha \cdot \mu_i^2$$

Parameter under frequency-severity model	Estimate
α	1.613293963434657

$\boldsymbol{\mu_i}$	$\widehat{\mu_i}$
n = 1/$\boldsymbol{\alpha}$	0.62984983683385
p	=1/(1 + $\alpha \cdot \mu_i$)
$\boldsymbol{\widehat{\mu}}$	6.827584815579126
$\boldsymbol{\widehat{\sigma}^2}$	1.1152302763020694

Table4. Parameter of simulated frequency-severity distribution

Here μ_i is the fitted mean claim frequency from the GLM (with exposure incorporated via an offset), and α is the dispersion parameter capturing unobserved heterogeneity.

For Severity Model, Conditional on claim occurrence, individual severities are independently simulated from the fitted lognormal distribution:

$$X_{ik} \sim Lognormal(\hat{\mu},\ \hat{\sigma}^2)$$

3.3 Simulation Algorithm

We generate S = 10000 simulated policy years. In each simulation s=1,2…S year.

1. **Simulate frequency per policy.**
   Draw $N_i^{(s)} \sim NB(\mu_i, \alpha)$ independently across policies (conditional on covariates).
2. **Aggregate total claim count.**
   Compute $N_{tot}^{(s)} = \sum i\ N_i^{(s)}$.
3. **Simulate severities using lognormal distribution.**
   If $N_{tot}^{(s)} > 0$, draw $N_{tot}^{(s)}$ severities from $X_{ik} \sim Lognormal(\hat{\mu},\ \hat{\sigma}^2)$ and sum:

$$L^{(s)} = \sum_{k=1}^{N_{tot}^{s}} X_k^{(s)}$$

Otherwise, set $L^{(s)} = 0$.

This produces an empirical distribution $\{L^{(1)}, \dots, L^{(S)}\}$ approximating the portfolio loss distribution.

Baseline Results & Analysis

Baseline Results

After S=10000 Monte Carlo simulations, an empirical distribution of aggregate losses is obtained. This simulated distribution shown below provides an approximation to the portfolio-level loss distribution and allows for a detailed examination of tail behavior and risk characteristics.

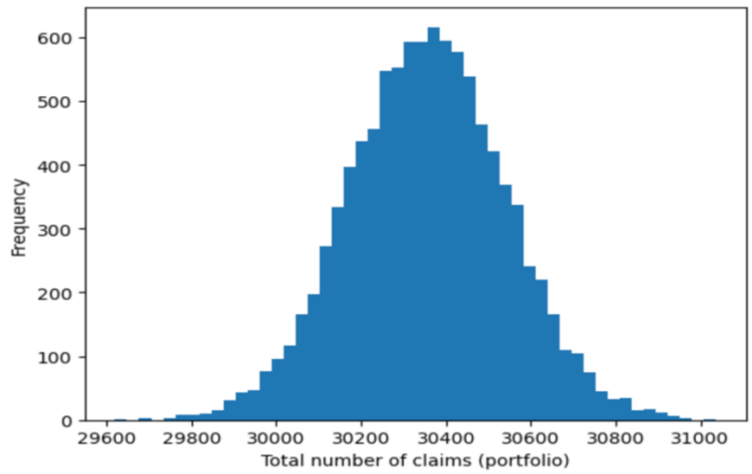

Figure 5.Distribution of Losses of Simulated Portfolio

From the simulated loss distribution, we derive the expected loss (EL), Value-at-Risk (VaR) and Expected Shortfall (ES) in one year period. According to the OSFI, it regulated 99% of the expected shortfall (conditional tail expectation) over a one-year time horizon. (OSFI,2024) Then we calculated the 99% quantile of the loss and divide them with the annual total exposure which derives from the dataset.

$$EL = \frac{1}{S} \cdot \sum_{s=1}^{S} L^{(s)}, VaR(k) = \inf\{x | F_x(x) \geq k\}, ES_k(x) = E[x | x > VaR]$$

Parameter	Estimate
EL per exposure	225.1066
VaR (99%) per exposure	230.7577
ES (99%) per exposure	231.4302

Table 5. Expected Value & Tail Behavior of simulated loss distribution.

Model Validation

To assess the stability of the pricing framework, the dataset is randomly split into a training sample (70%) and a testing sample (30%). The frequency model is estimated on the training sample, and expected loss per unit of exposure is evaluated on both samples using the same fitted parameters.

The resulting expected loss estimates are highly consistent across the training and testing datasets, indicating that aggregate loss estimates are stable and not driven by sample-specific variation. This suggests that the model provides a robust representation of portfolio-level risk rather than overfitting to a particular subset of observations.

Dataset	EL per exposure
Train dataset	225.1820
Test dataset	225.4970

Table 6. EL estimated by train & test dataset

This validation confirms that conclusions drawn from the model are robust at the portfolio level.

Risk Load and Final Premium

Although the pure premium represents the expected loss, it does not account for the variability and tail risk of the loss distribution. In insurance practice, capital must be held to absorb adverse deviations from expected outcomes.

To quantify this risk, VaR and ES we get from Monte Carlo simulation can be used in the process of quantifying risks. A capital-based pricing approach is therefore adopted, where the premium consists of the expected loss plus a risk load reflecting the cost of holding capital against extreme loss scenarios.

$$Premium = E[L] + Risk\ load$$

The capital required to support underwriting risk is ultimately provided by the insurer's capital providers, primarily shareholders. Holding capital against tail loss scenarios entails an opportunity cost, as this capital could otherwise be deployed for alternative investments or returned to investors. Therefore, a risk load is incorporated into pricing to compensate capital providers for bearing underwriting risk.

To translate tail risk into a pricing adjustment, an explicit assumption on the cost of capital is required.

Based on industry statistics, the average cost of capital for insurance-related firms is approximately **7%**, which is adopted in this study as an external benchmark rather than an internally estimated parameter.(KPMG, n.d.)

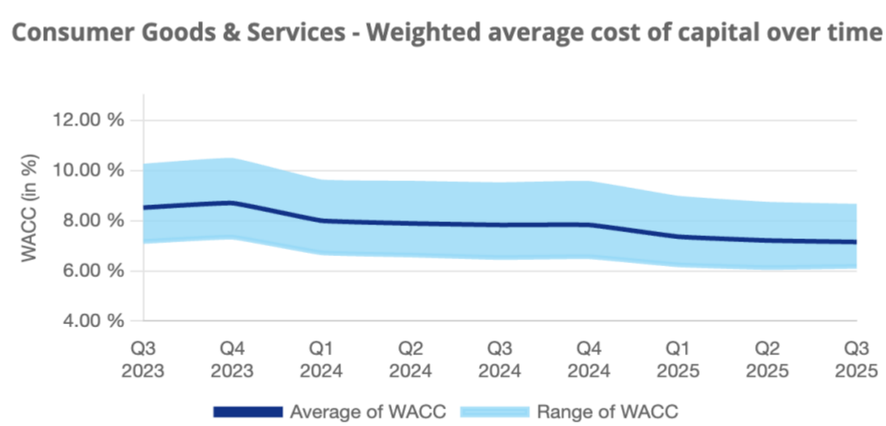

Figure 6. average cost of capital by KPMG

Then we get that

$$Risk\ Load = r_{wacc} \cdot (VaR\ per\ exposure - EL\ per\ exposure), \qquad r_{wacc} = 7\%.$$

Under the assumed capital cost and simulated loss distribution, the resulting premium is approxi mately: 225.5022 per unit of exposure.
This value should be interpreted as an **illustrative benchmark** rather than an operational tariff, a s it depends on externally specified capital cost assumptions.

EV Transition Stress Test

1.Motivation

The global transition toward electric vehicles (EVs) represents a structural shift in the motor insurance landscape. Compared to traditional internal combustion engine (ICE) vehicles, EVs are associated with higher repair complexity, driven by battery replacement costs, advanced sensor systems, and specialized labor requirements.
As a result, pricing models calibrated on historical ICE-dominated portfolios may underestimate future loss levels and tail risk. This makes EV transition a relevant emerging risk for both insurance pricing and capital adequacy assessment.
When analyzing the dataset, we find the datasets are composed of ICE vehicles. In this case, it's meaningful for invest how the occurrence of EVs will shock the current premium structure.

2.Stress Testing

It is important to emphasize that the EV transition analysis conducted in this section is **not a predictive model**.

Due to the absence of explicit EV indicators in the dataset, such as vehicle type, battery characteristics, or autonomous driving features, it is neither feasible nor appropriate to perform policy-level EV prediction. Instead, the analysis adopts a **stress testing perspective**, focusing on the robustness of the pricing framework under plausible structural shifts in loss dynamics.

The purpose of the stress test is therefore to evaluate **pricing adequacy under adverse but realistic scenarios**, rather than to forecast the exact future loss distribution.

3. Stress Scenario Design

Frequency Shock

According to the Guy Carpenter's report "Analysis of Electric Vehicles (EV) Accidents", EV accident frequency is consistently lower than that of ICEVs, with an overall reduction of 17% across all years tested in our analysis. Then, the baseline frequency distribution is scaled by a constant factor -17% to reflect the decrease of claims. (Guy Carpenter, 2025)

Severity Shock

Empirical studies and industry reports suggest that EV transition may alter not only the mean level of claim severity but also its tail behavior, due to higher uncertainty in battery damage and repair costs. The baseline severity distribution is scaled by a multiplicative factor to reflect increased repair and replacement costs. The following stress levels are considered:

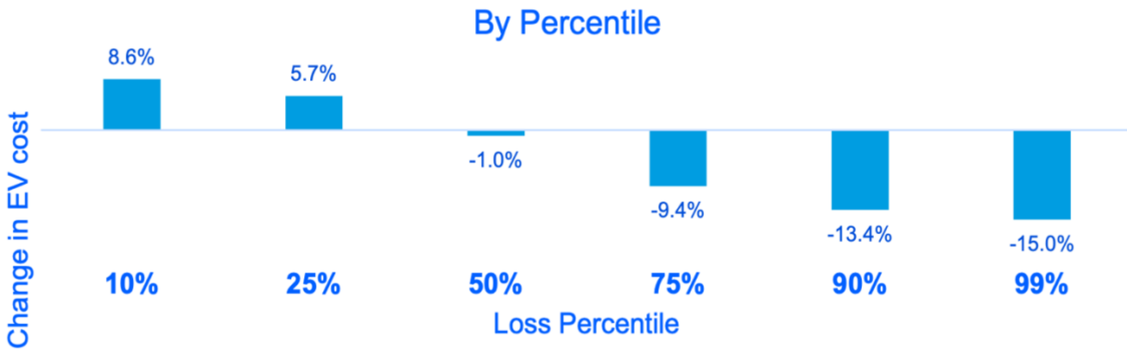

Figure 7.Change in EV cost by Guy Carpenter, 2025

These stress levels are designed to capture a range of plausible outcomes as EV adoption increases over time. (Guy Carpenter, 2025)

This scenario is not intended to represent a calibrated EV loss model, but rather to assess the sensitivity of capital requirements to potential changes in tail risk under regime transition.

4.Impact on Risk Load and Capital Requirement

Under the EV stress assumptions, we applied the Monte Carlo Simulation again and derive the following table:

	Pure premium(EL)	Risk Load	Premium
Baseline model	225.1066	0.3956	225.5022
Model under EV stress	168.6741	0.3196	168.9937
ChangeΔ	-66.4325	-0.076	-56.5085 (-25%)

Table 7. Change of pure premium, risk load and premium under stress

Implication and Discussion

Interpretation of EV Stress Test Results

Due to data limitations, this study does not attempt to empirically estimate EV-specific loss distributions. All conclusions regarding potential risk underestimation should therefore be interpreted as **model-based insights** rather than empirical forecasts.

The table shows that under the EV transition stress scenario, the model produces a **lower premium level** relative to the baseline, driven primarily by the assumed reduction in claim frequency outweighing the upward severity shock.

Although the EV stress scenario leads to a lower premium under the current assumptions, this does not necessarily imply a reduction in long-term risk.

Instead, the result reflects the dominance of frequency reduction in the modeled loss distribution, while potential increases in tail risk in severity beyond proportional scaling, parameter uncertainty given to the limited data, and model risk associated with applying ICE-calibrated models to structurally different vehicles.

As a result, capital-based pricing frameworks calibrated solely on adjusted mean parameters may understate the true risk compensation required during periods of structural transition.

The "Pricing Trap": Divergence between Model and Market

Our stress test suggests a potential premium reduction (~25%) driven by the lower accident frequency of EVs. However, this creates a significant contrast with current market realities.

According to The Post, premiums for EV are around 41% higher than for ICE, showing a totally opposite trend as our model. (The Post, 2023) As a result, a capital-based pricing framework calibrated solely on adjusted mean parameters may **underestimate the compensation required for tail risk during periods of structural transition**, even if the modeled premium decreases in the short run.

This divergence highlights a critical "Pricing Trap" for insurers:

- **Technical vs. Operational Risk:** Our model captures the *theoretical* frequency/severity trade-off. However, it does not capture operational frictions currently plaguing the EV repair ecosystem, such as **longer repair cycle times** (leading to higher replacement vehicle costs) and **scarcity of qualified high-voltage mechanics**.
- **Uncertainty Loading:** The market pricing reflects a substantial "ambiguity premium." Insurers are likely charging a heavy risk load not just for the *expected* tail risk (which we modeled), but for the **unmodeled uncertainty** surrounding battery degradation, supply chain volatility, and potential latent defects.
- **Conclusion:** An insurer relying solely on a GLM-based frequency discount (as simulated) would likely be **severely underpriced**. The model result serves as a "pure risk" baseline, while the gap to market price represents the cost of operational inefficiency and epistemic uncertainty.

Limitations

This study is subject to several limitations:

1. **Absence of EV-specific policy information**
   The dataset does not contain explicit indicators for vehicle electrification, battery characteristics, or autonomous features. EV effects are therefore introduced exogenously via stress scenarios rather than endogenously estimated.

2. **Simplified severity shock structure**
   Severity changes are modeled through proportional scaling of historical claims, which may not fully capture changes in tail shape or claim composition under EV technology.
3. **Intercept-only lognormal severity modeling**
   Given the high variability and weak dependence of severity on observed factors, an intercept-only Lognormal model was chosen. This approach prioritizes capturing the global volatility and tail structure of the portfolio over introducing noise from statistically insignificant predictors.
4. **Exclusion of non-insurance risks**
   Market, operational, regulatory, and legal risks associated with EV transition are not included in the process of modelling.
5. **Excluding Gas Type in the Model**
   By excluding gas type, the baseline model may underrepresent heterogeneity in driving patterns associated with different fuel types. However, this trade-off is accepted to preserve structural consistency when extending the model to electric vehicle (EV) scenarios.

Conclusion

This project develops a **solvency-based pricing framework** that links frequency and severity modeling with Monte Carlo simulation to quantify tail risk and translate it into a capital-based risk load.

The analysis demonstrates that:

- Expected loss alone is insufficient for pricing under uncertainty.
- Tail risk measures such as VaR and ES provide economically meaningful signals for capital allocation
- Stress testing offers a powerful tool for evaluating pricing robustness under structural change

Applying the framework to EV transition scenarios illustrates how emerging technologies can reshape loss dynamics in ways not captured by historical data alone. Even when expected premiums decline, latent tail risks and model uncertainty may warrant continued prudence in capital and pricing decisions.

Overall, the framework highlights the importance of **risk-aware pricing, capital discipline, and scenario-based thinking** in an evolving insurance landscape.

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Appendix A: Variable Definitions and Grouping

Variable	**Definition**
ClaimAmount	The amount of claims if happened
ClaimNb	The number of claims during the exposure period.
Exposure	The period of exposure for a policy, in years.
Power	The power of the car (split into ordinal categories).
CarAge	The vehicle age, in years.
DriverAge	The driver's age, in years (in France, people can drive a car from the age of 18).
Brand	The brand of the car
Gas	The car's fuel type: Diesel or Regular (petrol?).
Region	The policy region in France (based on the 1970-2015 region classification).
Density	The density of inhabitants, in the city that the driver of the car lives in, in (number of inhabitants per km^2).

Variable Definition Table

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
ClaimNb	0.00	0.00	0.00	0.039163	0.00	4
Exposure	0.002732	0.20	0.54	0.561088	1.00	1.99
CarAge	0.00	3.00	7.00	7.532404	12.00	100.00
DriverAge	18.00	34.00	44.00	45.319876	54.00	99.00
Density	2.00	67.00	287.00	1985.153913	1410.00	27000.00

Policy-level Descriptive Statistics（Frequency dataset）

Variable	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
ClaimAmount	2.00	698.00	1156.00	2129.972	1243.00	2036833.00
CarAge	0.00	3.00	7.00	7.667697	11.00	99.00
DriverAge	18.00	33.00	44.00	44.862431	54.00	99.00
Density	2.00	84.00	388.00	2113.642606	1466.00	27000.00

Claim-level Descriptive Statistics（Severity dataset）

Variable	Band	Range(year)	Reason
DriverAge	Young	18-25	Young drivers typically exhibit higher claim frequency due to limited driving experience
	Early Mid-age	26-35	Transition to more stable driving behavior
	Mid-age	36-45	Lower observed claim frequency and mature driving patterns
	Late Mid-age	46-55	Stable risk with slight increase in exposure
	Senior	56-65	Gradual increase in risk due to age-related factors
	Elderly	66+	Higher risk associated with declining reaction time

Driver age Band Table

Variable	Band	Range(year)	Reason
CarAge	New	0-2	Newer vehicles generally have lower mechanical-related risk and different repair patterns
	Recent	3-5	Stable risk segment with sufficient exposure
	Mature	6-10	Vehicle aging may increase accident/maintenance-related claim frequency
	Old	11+	Older vehicles may exhibit higher risk due to wear and safety feature gaps; extreme ages grouped for robustness.

Car age Band Table

Variable	Band	Range	Reason
Density	Rural/Low density	1-200	Low traffic intensity and exposure
	Semi-Urban	201-1000	Moderate congestion and collision opportunities
	Urban	1001-2000	Higher traffic density, increased claim frequency

	Higher Urban	2001-5000	Strong congestion effects and higher accident risk
	Very High Urban	5001+	Extremely dense areas; grouped to retain credibility in the tail

Density Band Table

Variable	**Group**	**Definition**	**Reason**
Power	D	Original power category d	Retained due to sufficient exposure and distinct risk level
	E	Original power category e	Retained due to sufficient exposure and interpretability
	F	Original power category f	Higher performance segment with adequate credibility
	G	Original power category g	Higher performance segment with adequate credibility
	Higher	Original categories above g	High-power vehicles consolidated to ensure statistical stability

Power Group Table

(Vehicle power categories with limited exposure were consolidated into a single "Higher" group to ensure parameter stability and avoid overfitting.
Power categories d–g were retained as separate rating factors due to sufficient exposure and meaningful differentiation in risk characteristics.)

Variable	**Group**	**Definition**	**Reason**
Region	Low Exposure Regions	{Basse-Normandie, Haute-Normandie, Limousin}	Consolidated due to low exposure to improve credibility and parameter stability
	Other Regions	All remaining regions kept as-is	Retained due to sufficient exposure and interpretability

Region Group Table

(Regions with limited exposure were consolidated into a single LowExposureRegions category to improve statistical credibility and stabilize parameter estimates.
Remaining regions were retained as separate levels to preserve geographic differentiation in claim frequency.)

Appendix B: Full Model Output（改）

Variable	**Coef.**	**95% CI**	**p-value**
Intercept(Baseline)	-1.8848	[-2.011, -1.759]	<0.001
DriverAge Band(reference 18-25）			
26-35	-0.7048	[-0.768, -0.641]	<0.001
36-45	-0.7610	[-0.824, -0.698]	<0.001

46-55	-0.7197	[-0.782, -0.658]	<0.001
56-65	- 0.8615	[-0.932, -0.791]	<0.001
66+	- 0.9017	[-0.975, -0.829]	<0.001
CarAge Band (reference 0-2)			
3-5	-0.0195	[-0.072, 0.033]	0.468
6-10	0.0241	[-0.026, 0.074]	0.347
11+	-0.1311	[-0.182, -0.080]	<0.001
Density Band(reference 1-200)			
201-1000	0.1912	[0.149, 0.233]	<0.001
1001-2000	0.3689	[0.315,0.423]	<0.001
2001-5000	0.4405	[0.387, 0.495]	<0.001
5001+	0.4313	[0.347, 0.516]	<0.001
Power Group(reference higher power)			
D	-0.2226	[-0.280, -0.165]	<0.001
E	-0.1053	[-0.160, -0.051]	<0.001
F	-0.0592	[-0.111, -0.007]	0.025
G	-0.1171	[-0.169, -0.065]	<0.001
Region Group(reference baseline region)			
Bretagne	-0.0686	[-0.149, 0.011]	0.093
Centre	-0.0832	[-0.153, -0.013]	0.019
Ile-de-France	0.0170	[-0.068, 0.102]	0.694
Nord–Pas–de–Calais	0.0158	[-0.110, 0.078]	0.741
Pays-de-la-Loire	-0.026	[-0.077, 0.108]	0.741
Poitou-Charentes	0.034	[-0.107, 0.056]	0.542
LowExposureRegion	-0.017	[-0.061, 0.133]	0.472
Brand(reference fiat)			
Japanese(except Nissan) or Korean	-0.2569	[-0.350, -0.164]	<0.001
Mercedes, Chrysler or BMW	0.0374	[-0.070, 0.144]	0.494
Opel, General Motors or Ford	0.0575	[-0.034, 0.149]	0.218
Renault, Nissan or Citroen	-0.0510	[-0.131, 0.029]	0.213
Volkswagen, Audi, Skoda or Seat	0.0249	[-0.069, 0.119]	0.604
other	-0.0425	[-0.170, 0.085]	0.521

Table: Negative Binomial GLM – Claim Frequency

Quantity	**Estimate**
Number of Claims	16181
Mean of log(Claim Amount)	6828
Standard Deviation of log(Claim Amount)	2.53
Kurtosis(log-scale)	6.42

Table: Lognormal Severity Model